Technical Report

Dual-stack (IPv4 and IPv6) network design and implementation

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Module: CMPG 325, North-West University

Project ID: CMPG325-2026-124

Client ID: CLI-124

Client: Tlou Amusement Arcade & Family Fun Centre (Vryburg)

Assigned challenge: IPv6 dual-stack addressing and routing

Project details

Item	Detail
Student	Selahle, Lebo (47925019)
Project ID / Client ID	CMPG325-2026-124 / CLI-124
Client	Tlou Amusement Arcade & Family Fun Centre (Vryburg), Entertainment
Assigned block	192.168.54.0/24
Assigned challenge	IPv6 (dual-stack addressing & routing)
Design constraint	Internet bandwidth is limited and expensive, so the design must minimise waste
Change request	CR8: a shared printer zone must serve two departments that currently cannot print
Submission date	16 October 2026
Repository	https://github.com/SelahleLebo04/Tlou-Amusement-Arcade-Family-Fun-Centre-Network-

1.Introduction

This report documents the design, implementation, testing and troubleshooting of a computer network for Tlou Amusement Arcade & Family Fun Centre, a small single-site venue with an arcade floor, a cafe and a back office. Cisco Packet Tracer was used to build and simulate the network. The client change request CR8 adds a shared printer zone for the Arcade and Cafe departments, and IPv6 dual-stack addressing and routing is the designated technical difficulty. The requirements, design, IP addressing, Packet Tracer implementation, the designated feature, testing, and troubleshooting are all included in the report. The GitHub portfolio has all of the evidence mentioned here.

2 Client Requirements

Since the client is a small business, the network needs to be easy to maintain, inexpensive to operate, and robust enough that a single malfunction won't stop ticketing, POS, or the arcade floor.

Group	Function	Devices	Network need
Administration	Manager PC, bookkeeping, rostering, back-office printing	5	Most sensitive data; least exposure

Group	Function	Devices	Network need
Arcade floor	Redemption/POS kiosks, machine controllers	20	Reliable LAN, low internet use; needs printing (CR8)
Cafe & snack bar	POS tills, kitchen display	8	Reliable LAN, low internet use; needs printing (CR8)
Guest Wi-Fi	Patron phones and tablets	40 concurrent	Internet only; isolated from internal systems
Shared services	Internal DHCP/DNS server, switch management	5	Local services that do not depend on the WAN
Printer zone (CR8)	Shared printer(s) for Arcade and Cafe	4	Reachable only by Arcade and Cafe

Security requirements. Guest Wi-Fi must be logically isolated from POS, admin and printer traffic. The printer zone must be reachable only from Arcade and Cafe. The Admin network holds the most sensitive data.

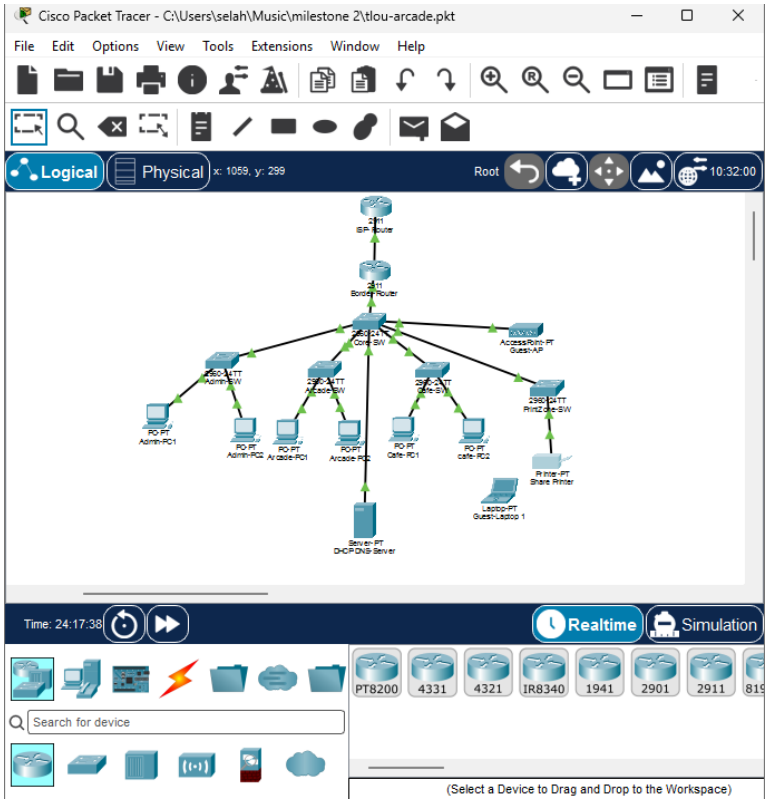
Budget and bandwidth. One appropriately sized WAN link, local DHCP and DNS, POS and arcade traffic kept on the LAN, no cloud-dependent services, and Guest Wi-Fi as the only group expected to use significant internet bandwidth.

Growth. About 40% of the /24 is left unallocated so more machines or a new kiosk area can be added without renumbering.

3 Network Design

3.1 Physical topology

A single border router connects the ISP to a core switch. The core switch trunks to the Admin, Arcade and Cafe access switches and to a print-zone switch, and connects the Guest access point and the DHCP/DNS server directly. A single router and core switch keeps hardware cost low, while VLANs provide logical separation between departments without a separate physical switch for each one.



3.2 Logical design (VLANs)

VLAN	Name	Purpose	Access policy
10	ADMIN	Back-office PCs	Internal; NAT to internet
20	ARCADE	Kiosk/POS terminals	Internal; can use the printer zone; no IPv4 NAT
30	CAFE	Cafe POS terminals	Internal; can use the printer zone; no IPv4 NAT
40	GUEST	Patron Wi-Fi	Isolated from Admin, Services and Print-zone; NAT to internet
50	PRINT-ZONE	Shared printer (CR8)	Reachable only from Arcade and Cafe
60	SERVICES	DHCP/DNS server, switch management	Internal
99	WAN	Link to the ISP (192.168.54.144/30)	Point-to-point

The border router uses router-on-a-stick to do inter-VLAN routing. One trunk (GigabitEthernet0/0) contains 802.1Q sub-interfaces.10 to.60, each of which serves as the VLAN's default gateway. This lowers costs and provides a single point of access management.

3.3 IPv4 addressing (VLSM)

The /24 was divided by VLSM, largest subnet first, so each VLAN gets only the space it needs. About 108 addresses remain for growth.

VLAN	Subnet	Gateway	Hosts	Usable range
40 GUEST	192.168.54.0/26	.1	40 of 62	.1 – .62
20 ARCADE	192.168.54.64/27	.65	20 of 30	.65 – .94
30 CAFE	192.168.54.96/28	.97	8 of 14	.97 – .110
10 ADMIN	192.168.54.112/28	.113	5 of 14	.113 – .126
60 SERVICES	192.168.54.128/29	.129	5 of 6	.129 – .134
50 PRINT-ZONE	192.168.54.136/29	.137	4 of 6	.137 – .142
99 WAN	192.168.54.144/30	.145 / .146	2 of 2	.145 – .146

3.4 IPv6 addressing

No ISP-delegated IPv6 prefix is available in the lab, so a locally assigned Unique Local Address (ULA) prefix, FD54:ACDE:2026::/48, is used. Each VLAN receives a /64, which is what SLAAC requires, and the router takes ::1 in each subnet.

VLAN	IPv6 subnet	Assignment method
10 ADMIN	FD54:ACDE:2026:10::/64	SLAAC + stateless DHCPv6 (DNS)
20 ARCADE	FD54:ACDE:2026:20::/64	SLAAC + stateless DHCPv6 (DNS)
30 CAFE	FD54:ACDE:2026:30::/64	SLAAC + stateless DHCPv6 (DNS)
40 GUEST	FD54:ACDE:2026:40::/64	SLAAC only (isolated from the internal DNS server)
50 PRINT-ZONE	FD54:ACDE:2026:50::/64	Static (printer ::2)
60 SERVICES	FD54:ACDE:2026:60::/64	Static (server ::10, router ::1)
WAN	FD54:ACDE:2026:FFFF::/64	Static (router ::1, ISP ::2)

3.5 Design decisions and justification

- **VLANs instead of a flat network:** keeps POS and financial traffic away from patron devices, limits broadcast domains for about 80 concurrent devices, and gives the router a point to apply access control.
- **Router-on-a-stick:** one router handles inter-VLAN routing, NAT and ACLs without the cost of a Layer 3 switch or a second router.
- **VLSM:** an equal split across seven VLANs would waste addresses on small VLANs and risk running short for Guest and Arcade.
- **Dual-stack rather than tunnelling:** both protocols run natively on every sub-interface, so no tunnel overhead is added to the single WAN link.
- **Local DHCP/DNS:** the internal server leases addresses and resolves names without using the WAN.
- **NAT only for Guest and Admin:** the NAT list permits the Guest and Admin IPv4 subnets only, so Arcade and Cafe IPv4 traffic stays on the LAN, as the bandwidth constraint requires.

4.1 Devices

Device	Model	Role
BORDER-R1	Cisco 2911	Border router: inter-VLAN routing, dual-stack, NAT, ACLs, DHCPv6, DHCP relay
ISP	Cisco 2911	Simulated ISP with IPv4 loopback 8.8.8.8 and IPv6 loopback 2001:DB8:8::8
CORE-SW	Cisco 2960	Core switch: trunks to the router and access switches; Guest AP and server ports
ADMIN-SW / ARCADE-SW / CAFE-SW	Cisco 2960	Access switches for VLANs 10, 20 and 30
PRINTZONESW	Cisco 2960	Print-zone switch for the shared printer (VLAN 50)
DHCP-DNS-Server	Server-PT	Static 192.168.54.130/29 and FD54:ACDE:2026:60::10/64; DHCP and DNS
Access point / Guest-Laptop 1	AP-PT / Laptop-PT	Guest Wi-Fi in VLAN 40

Device	Model	Role
Shared Printer	Printer-PT	Static 192.168.54.138/29 and FD54:ACDE:2026:50::2/64
PCs	PC-PT	Two each for Admin, Arcade and Cafe, using DHCP and automatic IPv6

4.2 Layer 2 configuration

VLANs 10 to 60 exist on the switches, and each access switch places its host ports in its own VLAN and has a management address in VLAN 60 with default gateway 192.168.54.129. CORE-SW trunks to BORDER-R1 on Fa0/1 (VLANs 10 to 60 allowed) and to the three access switches on Fa0/2 to Fa0/4. The Guest access point is on Fa0/5 (VLAN 40), the print-zone switch on Fa0/6 (VLAN 50) and the server on Fa0/7 (VLAN 60).

Figure 2: ADMIN-SW: ports in VLAN 10, management address in VLAN 60 and ping to the gateway.

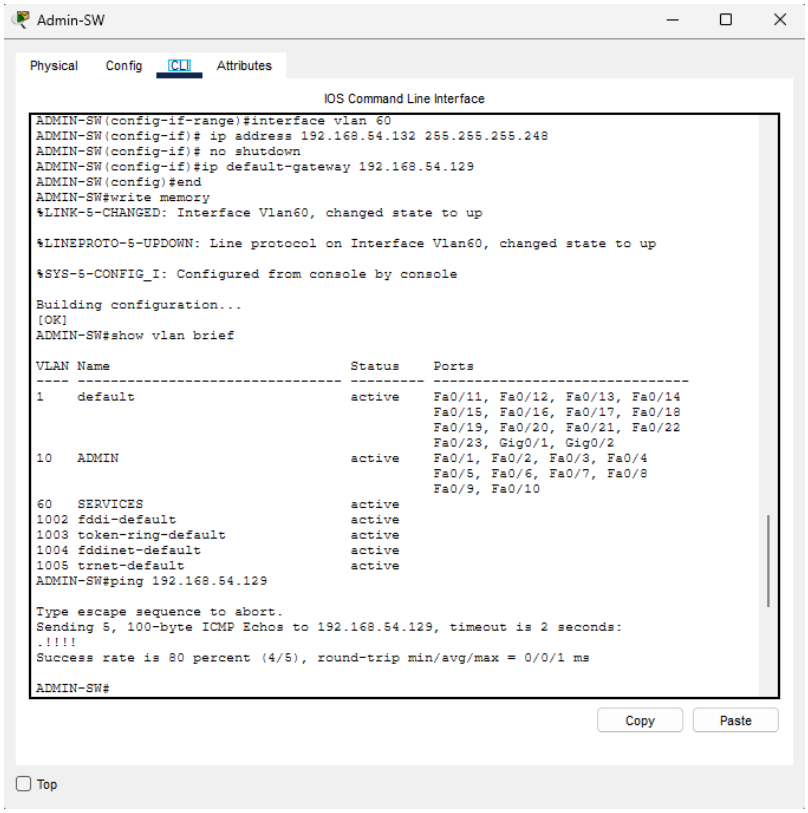


Figure 3: ARCADE-SW: ports in VLAN 20 and ping to the gateway

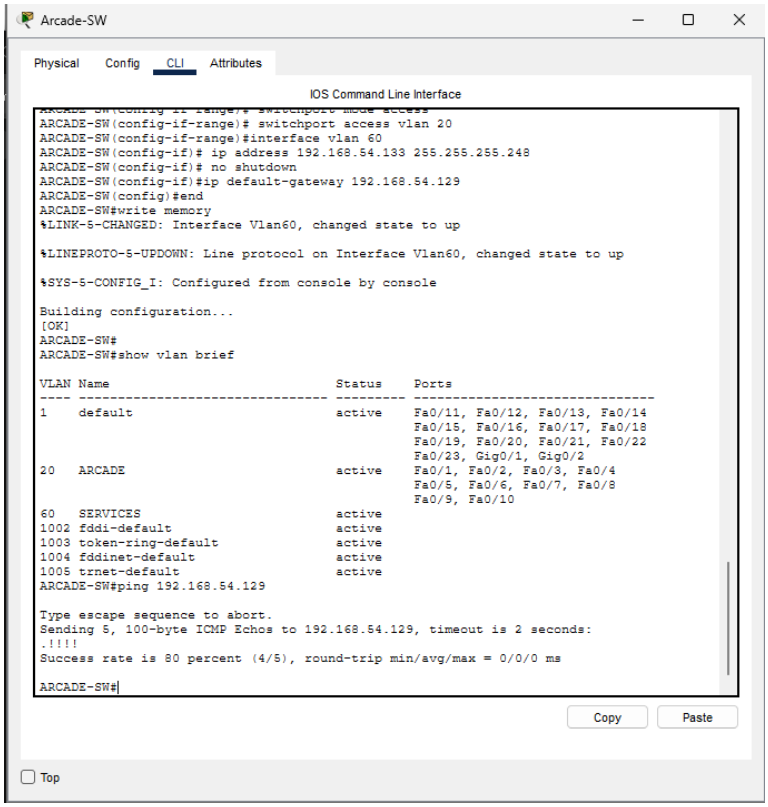


Figure 4: CAFE-SW: ports in VLAN 30 and ping to the gateway

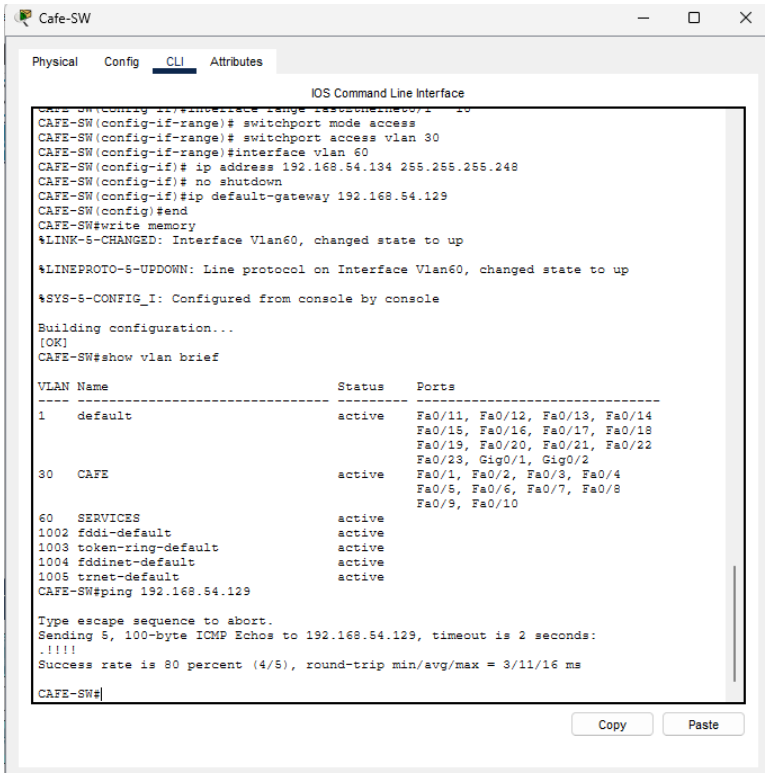
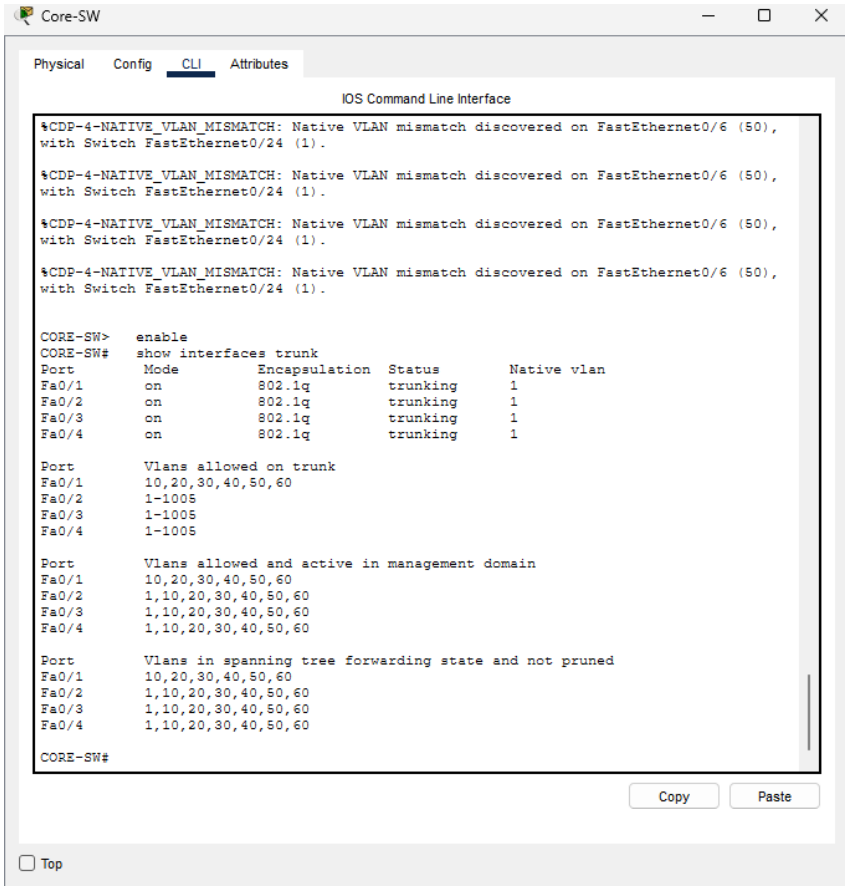


Figure 5: CORE-SW show interfaces trunk: Fa0/1 carries VLANs 10 to 60 to the router and Fa0/2 to Fa0/4 trunk to the access switches



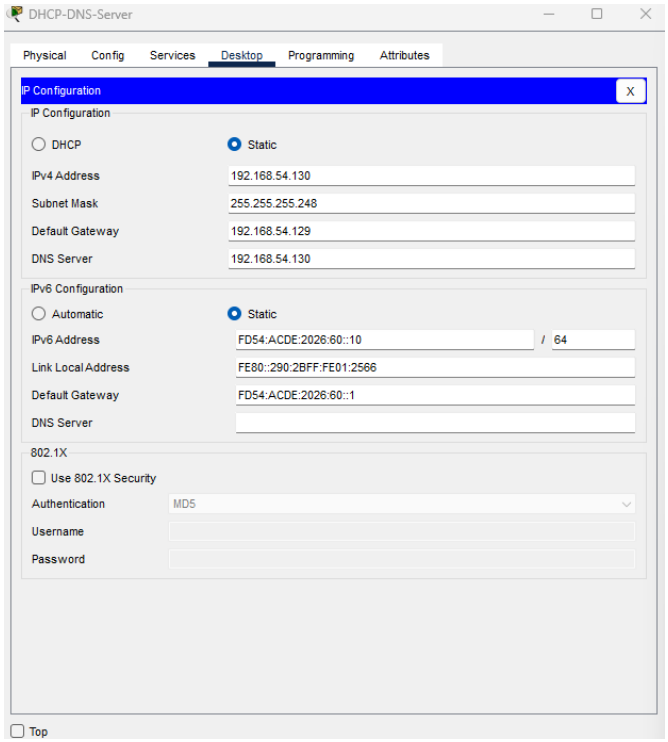
4.3 Router, DHCP and NAT

BORDER-R1 carries six 802.1Q sub-interfaces on GigabitEthernet0/0 and the WAN on GigabitEthernet0/1. Guest, Admin, Arcade and Cafe sub-interfaces use ip helper-address 192.168.54.130 so DHCPv4 requests reach the internal server. NAT overload translates the permitted subnets to the WAN address 192.168.54.145, and default routes (IPv4 and IPv6) point to the ISP

.

```
interface GigabitEthernet0/0.20
 description ARCADE
 encapsulation dot1Q 20
 ip address 192.168.54.65 255.255.255.224
 ip helper-address 192.168.54.130
 ipv6 address FD54:ACDE:2026:20::1/64
 ipv6 nd other-config-flag
 ipv6 dhcp server LAN-V6
interface GigabitEthernet0/1
 description WAN to ISP
 ip address 192.168.54.145 255.255.255.252
 ip nat outside
 ipv6 address FD54:ACDE:2026:FFFF::1/64
ip nat inside source list 1 interface GigabitEthernet0/1 overload
ip route 0.0.0.0 0.0.0.0 192.168.54.146
ipv6 route ::/0 FD54:ACDE:2026:FFFF::2
access-list 1 permit 192.168.54.0 0.0.0.63
access-list 1 permit 192.168.54.112 0.0.0.15
```

Figure 6: DHCP-DNS-Server static addressing: 192.168.54.130/29 and FD54:ACDE:2026:60::10/64



5 Assigned Challenge: IPv6 Dual-Stack Addressing and Routing

5.1 What was configured

- ipv6 unicast-routing enabled on BORDER-R1 and on the ISP router, so both forward IPv6 packets.
- A global IPv6 address (::1 of a /64) on every router sub-interface and on the WAN interface, next to the existing IPv4 address, so both protocols run together on each interface.
- Router advertisements enable SLAAC: hosts build their own address from the /64 prefix. ipv6 nd other-config-flag tells hosts on VLANs 10, 20 and 30 to ask DHCPv6 for other information (DNS).
- Stateless DHCPv6 pool LAN-V6 provides DNS server FD54:ACDE:2026:60::10 and domain tlou.local, applied on VLANs 10, 20 and 30. Guest uses SLAAC only.
- Static IPv6 on the printer (FD54:ACDE:2026:50::2) and the server (FD54:ACDE:2026:60::10).
- A static IPv6 default route on the border router toward the ISP (FD54:ACDE:2026:FFFF::2) and a return route on the ISP for the whole FD54:ACDE:2026::/48 prefix.
- IPv6 access lists GUEST6-IN and TO-PRINTZONE6 enforce the same security rules as the IPv4 lists.

```
ipv6 unicast-routing
ipv6 dhcp pool LAN-V6
 dns-server FD54:ACDE:2026:60::10
 domain-name tlou.local
ipv6 access-list GUEST6-IN
 permit ipv6 any FD54:ACDE:2026:40::/64
 permit ipv6 any FD54:ACDE:2026:FFFF::/64
 deny ipv6 any FD54:ACDE:2026::/48
 permit ipv6 any any
```

5.2 Why this is appropriate

• Both protocols are run natively by dual-stack. It is the most straightforward migration route for a small organization and adds no overhead to the constrained WAN link because it doesn't require tunnelling or translation.

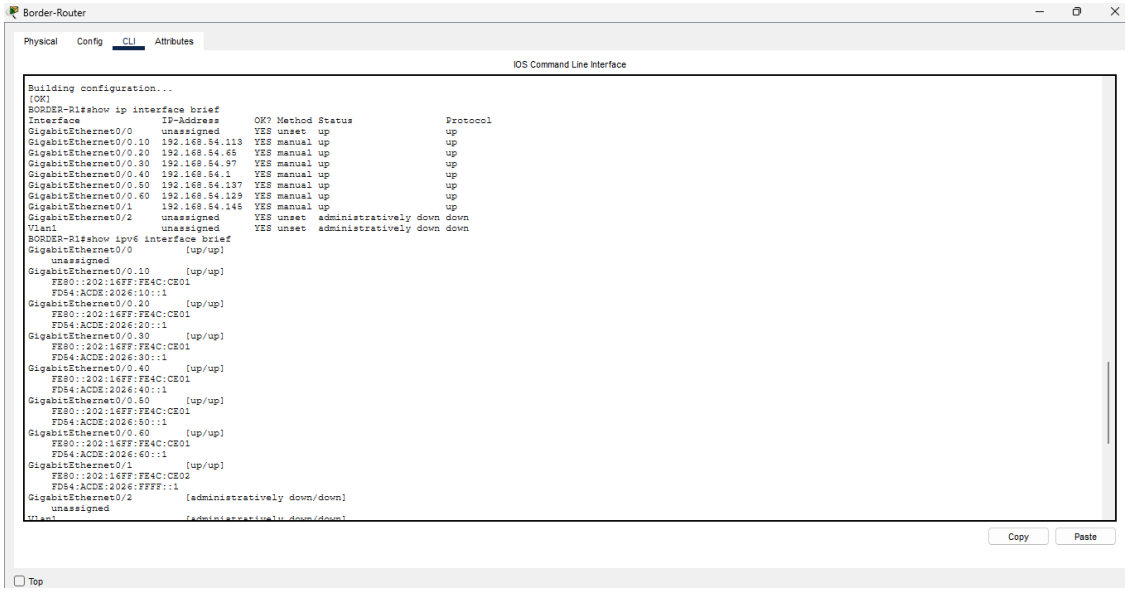
- A ULA prefix is correct for a lab with no ISP-delegated prefix. It can be replaced with a global prefix later by changing the prefix, because the /64-per-VLAN structure stays the same.

- SLAAC needs no per-host state on the router, and stateless DHCPv6 adds DNS without a DHCPv6 lease server, which suits a small site.
- Guest is SLAAC only because it is blocked from the internal DNS server, so a DNS address would be unreachable for guests.

5.3 How it was verified

Every sub-interface shows both an IPv4 address and an IPv6 link-local and global address, and is up/up:

Figure 7: BORDER-R1 show ip interface brief and show ipv6 interface brief: all sub-interfaces up with IPv4 and IPv6



Hosts receive a DHCPv4 lease and a SLAAC address, with the router link-local address as the IPv6 default gateway.

Figure 8: Admin-PC1: DHCP lease 192.168.54.114 and SLAAC address in FD54:ACDE:2026:10::/64

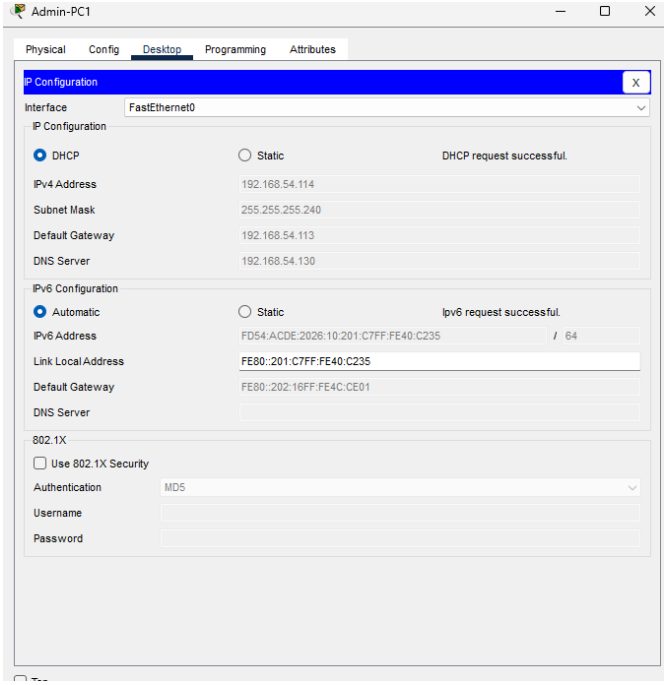


Figure 9: Arcade-PC1 ipconfig /all: lease from DHCP server 192.168.54.130 and SLAAC address in FD54:ACDE:2026:20::/64

```
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Physical Address.................: 0005.5E50.595A
   Link-local IPv6 Address.........: FE80::205:5EFF:FE50:595A
   IPv6 Address....................: FD54:ACDE:2026:20:205:5EFF:FE50:595A
   IPv4 Address....................: 192.168.54.66
   Subnet Mask.....................: 255.255.255.224
   Default Gateway.................: FE80::202:16FF:FE4C:CE01
                                     192.168.54.65
   DHCP Servers....................: 192.168.54.130
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-62-02-46-88-00-05-5E-50-59-5A
   DNS Servers.....................: ::
                                     192.168.54.130

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Physical Address.................: 0050.0F73.5B2C
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
   DHCP Servers....................: 0.0.0.0
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-62-02-46-88-00-05-5E-50-59-5A
   DNS Servers.....................: ::
                                     192.168.54.130


C:\>
```

IPv6 is routed between subnets and to the printer, and IPv6 security policy is enforced.

6 Change Request CR8: Shared Printer Zone

Arcade and Cafe previously had no printing. Rather than buying a printer for each department or placing a printer on the Admin VLAN, a dedicated PRINT-ZONE VLAN 50 was created. The ACLs TO-PRINTZONE (IPv4) and TO-PRINTZONE6 (IPv6) are applied outbound on the VLAN 50 sub-interface. They permit only the

Arcade and Cafe subnets to reach the printer, on the print ports (9100, 515, 631 for IPv4; 9100 for IPv6) and ICMP, and deny everything else. Admin and Guest are blocked. This satisfies the wording of CR8 without adding scope.

Figure 10: BORDER-R1 ping to the printer FD54:ACDE:2026:50::2: 5 of 5

```
BORDER-R1#ping FD54:ACDE:2026:50::2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to FD54:ACDE:2026:50::2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/2/11 ms

BORDER-R1#
```

7 Testing and Verification

ID	Test	Expected	Actual	Result
T1	Access switches place ports in the correct VLAN and reach 192.168.54.129	VLAN ports shown; ping succeeds	4 of 5 (first lost to ARP)	Pass
T2	Trunks on CORE-SW	Fa0/1 to Fa0/4 trunking; VLANs 10–60 allowed	As expected	Pass
T3	Router sub-interfaces dual-stack	IPv4 and IPv6 on every sub-interface, up/up	As expected	Pass
T4	Host DHCPv4 lease and SLAAC address	Lease from .130 and an IPv6 address in the VLAN /64	As expected	Pass
T5	Inter-VLAN IPv4: Admin-PC1 to server .130 and Arcade-PC1 .66	Replies	4 of 4 each	Pass
T6	CR8: Arcade to printer (IPv4 and IPv6)	Replies	4 of 4 each	Pass
T7	CR8: Cafe to printer (IPv4 and IPv6)	Replies	4 of 4 each	Pass

ID	Test	Expected	Actual	Result
T8.	Admin to printer	Blocked	Unreachable on both stacks	Pass
T9	Guest to Admin, Services, Print-zone (IPv4)	Blocked	Unreachable from 192.168.54.1	Pass
T10	Guest to ISP 192.168.54.146	Reachable	Replies, TTL 254	Pass
T11	Guest to its gateway and 8.8.8.8	Reachable	4 of 4 each	Pass
T12	translation for Guest NAT	192.168.54.2 translated to 192.168.54.145	ICMP entries present	pass
T13	Guest to Admin and Print-zone (IPv6)	Blocked	Unreachable from FD54:ACDE:2026:40::1	pass
T14	Guest to its own IPv6 gateway	Reachable	4 of 4	pass

8 Troubleshooting

1. **Guest could not reach the WAN.** The IPv4 list GUEST-IN contained deny ip any 192.168.54.128 0.0.0.127, which covers .128 to .255 and so also the WAN link. It was replaced with a 0.0.0.15 wildcard (.128 to .143, Services and Print-zone only). The first attempt, no 25, removed nothing because this Packet Tracer does not display sequence numbers. The old rule was removed by its text, and the corrected list was confirmed with show access-lists.
2. **Stray sub-interface and ACL.** GigabitEthernet0/1.40 and GUEST-OUT had been created on the WAN port instead of the trunk on Gi0/0. Both were removed.
3. **Printer unreachable over IPv6.** IPv4 worked, IPv6 failed. A ping from the router itself, which bypasses the print-zone ACL, also failed, so the ACL was ruled out. The printer had no static IPv6 address. Setting FD54:ACDE:2026:50::2/64 fixed it (router ping 5 of 5).

4. **Guest could not ping its own IPv6 gateway.** GUEST6-IN denied the whole /48. Permits for the Guest /64 and the WAN /64 were added before the deny.
5. **DHCPv6 missing and wrong DNS address.** The design specified stateless DHCPv6, but the router had no pool. Pool LAN-V6 and the other-config flag were added. The pool first pointed at ::2, so it was corrected to the server address ::10.
6. **Packet Tracer crashed.** The newest .pkt was reopened and the router configuration was intact. The last fixes were reapplied, and device configurations are now stored as text in the repository.

9 Limitations and Observations

- The DHCPv6 DNS option is configured on the router (show ipv6 dhcp pool) but the Packet Tracer PCs show a blank IPv6 DNS server. This is probably a Packet Tracer PC limitation. SLAAC addressing and routing are unaffected, and IPv4 DNS (192.168.54.130) is delivered by DHCPv4.
- IPv6 is not translated, so the NAT list that withholds IPv4 internet from Arcade and Cafe does not limit IPv6. In a real deployment an IPv6 access list would be added on the Arcade and Cafe sub-interfaces to enforce the same policy.
- CORE-SW reports a CDP native VLAN mismatch on Fa0/6, toward the print-zone switch. Traffic is unaffected and printing works, but the native VLAN settings should be aligned.
- The simulation uses two or three PCs per VLAN to represent the 20 kiosks, 8 tills and 40 guests, and a simulated ISP with loopback addresses stands in for the internet.
- Guest rate-limiting/QoS, suggested in the requirements, was not configured because it was outside the assigned challenge and CR8.

10 Reflection

Working on this project helped me understand how IPv4 and IPv6 work together and how to keep a network safe using VLANs and ACLs. I faced some problems while using Cisco Packet Tracer, especially with the wildcard mask and the printer's IPv6 address, but I managed to fix them by checking my configurations. When Packet Tracer crashed, I learned that it is important to save my work regularly so that I do not lose everything. I was happy when I finally got the network working and made sure that the Arcade and Cafe could use the printer while the other networks were blocked. Overall, this project taught me a lot about networking, and if I had more

time, I would improve the network security and find a better way to manage guest internet traffic.

11 Conclusion

The network meets the client requirements. All six VLANs route over IPv4 and IPv6, hosts obtain addresses automatically, the shared printer zone serves Arcade and Cafe only (CR8), Guest is isolated but has internet access through NAT, and POS and arcade traffic stays on the LAN to respect the bandwidth constraint. The IPv6 challenge was configured, explained and verified with the evidence above.

Appendix A: Repository contents

Location	Contents
README.md	Project overview, design tables and repository layout
docs/	Milestone 1 design documents, Milestone 2 report and this technical report
packet-tracer/tlou-network.pkt	The Packet Tracer simulation
configs/	Running configuration of each device (BORDER-R1, ISP, CORE-SW, ADMIN-SW, ARCADE-SW, CAFE-SW, PRINTZONESW)
testing/	connectivity-tests.md, troubleshooting.md and screenshots/
reflection.md	Project reflection